

Physics Factsheet



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Number 127

Molecules, Moles and Mass

This Factsheet focuses on questions and calculations about gases. This includes the Ideal gas equation and kinetic theory and covers molecules, moles, kinetic energy of a gas particle and rms velocity. There are several common mistakes which will be explained.

Ideal Gas Equation and the Mole

The equation of state for an ideal gas is given by $pV=nRT$, otherwise known as the Ideal Gas Equation. The pressure, p , the volume, V and the temperature, T are obvious from their symbols. The Molar Gas Constant, R is 8.31J/K/mol . The number of moles of gas, n , needs more explanation.

When you breathe in deeply, your lungs contain around 6 litres of air (depending on your size and other factors). These 6 litres of air consists of approximately 1.2×10^{24} individual particles: mostly molecules of nitrogen, oxygen and some carbon dioxide. A mole represents 6×10^{23} individual particles: either atoms or molecules.

This means our lungs contain around 2 moles of air, a much more down-to-earth figure. The exact number is based on carbon-12. If you have exactly 0.012kg of C-12, you have 1 mole of particles, which is actually 6.022×10^{23} particles and known as Avogadro's constant (L or N_A). One single atom of C-12 has a mass of $12u$ and one mole of C-12 atoms has a mass of 0.012kg .

 One mole is 6.022×10^{23} particles. This number of particles, atoms or molecules, is known as Avogadro's constant, N_A .

We can also consider molecules as well as atoms. This is more useful for gas calculations as gases are more commonly molecular and not atomic. One molecule of O_2 has a mass of $32u$. Therefore 1 mole of O_2 has a mass of 0.032kg .

Particle	Particle mass (u)	Mass of 1 mole
Carbon atom	12	0.012kg
Oxygen atom	16	0.016kg
Helium atom	4	0.004kg
O_2 molecule	32	0.032kg
CO_2 molecule	44	0.044kg
N_2 molecule	28	0.028kg

Exam Hint: In Chemistry, molar masses are given in grams. For Physics calculations, remember to keep all masses in kilograms to avoid mistakes.

Example 1

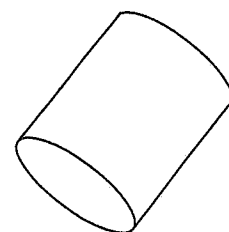
1. What is the atomic mass of Argon (in atomic mass units, not kg)?
2. What is the mass of 1 mole of Argon gas?
3. What is the mass of 4.5 moles of sulphur dioxide gas?
4. How many moles are there in 0.038kg of carbon monoxide molecules?

Answers

1. The atomic mass of Argon is 40 u (39.95).
2. One mole of Argon has a mass of 0.04kg .
3. The mass of a single SO_2 molecule = $32 + (2 \times 16) = 64u$
1 mole of SO_2 has a mass of 0.064kg
4.5 moles of SO_2 has a mass of $0.064\text{kg} \times 4.5 \text{ moles} = 0.288\text{kg}$
4. The mass of a single CO molecule = $12 + 16 = 28u$
1 mole of CO has a mass of 0.028kg
 $0.038\text{kg} / 0.028\text{kg} = 1.36 \text{ moles}$.

Ideal Gas Equation calculations

A sealed container holds carbon dioxide gas. The pressure inside is 10 atmospheres, the volume of the container is 200cm^3 and the temperature is 30°C . How many moles of gas are there in the container?



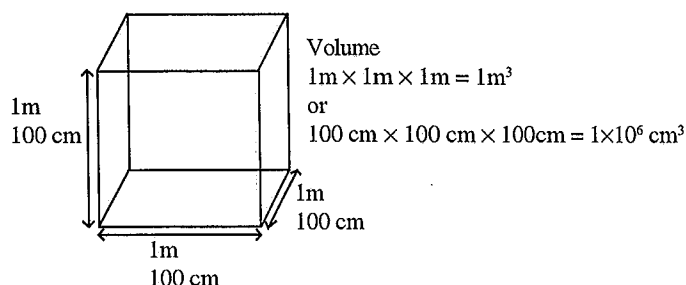
There are three common mistakes that could be made when answering this question:

$V = 200\text{ cm}^3$
 $T = 30^\circ\text{C}$
 $P = 10 \text{ atmospheres}$

Pressure must be in units of N/m^2 or Pascals. Volume must be in units of m^3 and the temperature must be in Kelvin on the absolute temperature scale.

Converting the pressure first: One atmosphere is equivalent to $1.013 \times 10^5 \text{ Nm}^{-2}$ or Pa. So $10 \times 1.013 \times 10^5 = 1.013 \times 10^6 \text{ Pa}$

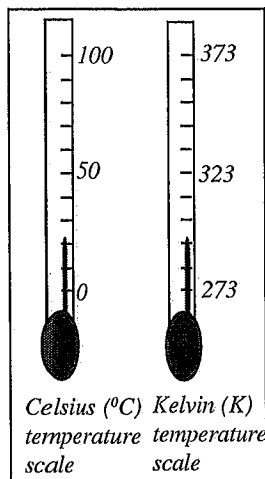
The volume must be in m^3 . How do we convert from cm^3 to m^3 ? Picture a box, with sides 1m by 1m by 1m . Each side is 100cm long. To find out how many cm^3 there are in the box, multiply 100cm by 100cm by $100\text{cm} = 1 \times 10^6 \text{ cm}^3$ or 1 million cm^3 . Volumes are often given in litres. 1 litre contains 1000cm^3 , so there are 1000 litres in 1m^3 .



So $200\text{cm}^3 / 1 \times 10^6 \text{ cm}^3 = 2 \times 10^{-4} \text{ m}^3$

Exam Hint: It is easy to make a mistake when converting from cm^3 to m^3 . If you have 200cm^3 , the result must be a **SMALL** fraction of a cubic metre.

Exam Hint: When using the Ideal Gas Equation the temperature must be in Kelvin, according to the absolute temperature scale. 0K is equivalent to -273.15°C . 0°C is equivalent to 273.15K . Add 273.15 to any temperature in $^\circ\text{C}$ to change it to Kelvin.



$$\text{So } 30^\circ\text{C} + 273.15 = 303.15\text{K}$$

$$pV = nRT$$

becomes $\frac{pV}{RT} = n$ after rearranging

$$\frac{1.013 \times 10^6 \text{ Pa} \times 2 \times 10^{-4} \text{ m}^3}{8.31 \times 303.15\text{K}}$$

$$= 0.080 \text{ moles.}$$

Example 2

1. How many molecules were in this sealed container?
2. What is the mass of the gas inside this sealed container?

Answers

1. $0.080 \text{ moles} \times 6.022 \times 10^{23} = 4.82 \times 10^{22} \text{ molecules.}$
2. One molecule of carbon dioxide has a mass of:
 $12 + 2 \times 16 = 44\text{u.}$ So one mole is 0.044kg
 $0.080 \text{ moles} \times 0.044\text{kg} = 3.52 \times 10^{-3}\text{kg}$

Kinetic Theory

Using the kinetic theory, we can derive a very useful equation:

$$pV = \frac{1}{3}Nm\overline{v^2}$$

This equation links large scale easy-to-measure quantities like pressure, p and volume, V to small scale quantities like the mass of a particle, m (in kg), number of individual particles, N , and average squared speed, $\overline{v^2}$.

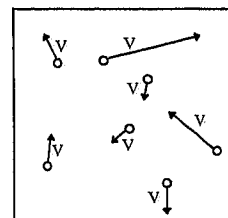
Exam Hint: The number of moles, n , and the number of individual particles, N , are **VERY** easy to confuse. One way to help remember the difference: n (lower case) will be a relatively small number, N (capital letter) will be a very large number.

Average squared speed is one concept that people sometimes struggle with. The N particles in any gas will be travelling with a range of speeds and every particle has a different velocity, v . This means that the squared speed for each particle, v^2 , is also different. For calculations involving many particles, we use an average of

this squared speed: $\overline{v^2}$, units m^2s^{-2} . (The reason we are more concerned with squared speed, rather than speed itself, is because squared speed is needed for kinetic energy calculations.)

Key: $\overline{v^2}$ is the average of the squared speeds of all of the particles in the gas. This is often known as the mean squared speed. Some textbooks

write this as $\overline{c^2}$.



Gas particles travel with a range of velocities

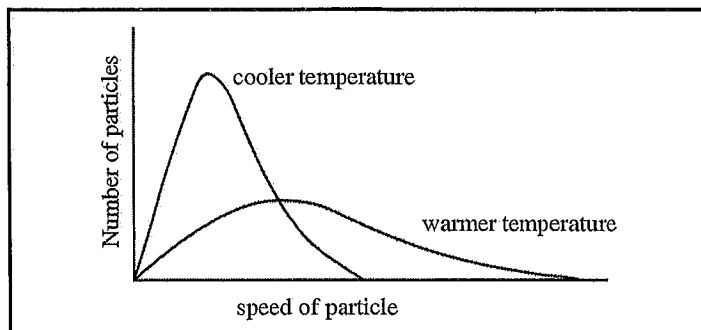
So, on average, how fast are the particles travelling?

Take the square root of the mean squared speed, $\sqrt{\overline{v^2}}$, this gives a good indication of how fast the particles travel on average.

$\sqrt{\overline{v^2}}$ is known as the root-mean-squared-velocity (or rms velocity).

A small proportion of the particles will be travelling very slowly; another small proportion will be travelling very fast. The majority of the particles will be travelling between these two extremes. If you look at the distribution of the particles across the different speeds, it forms a broad hump. This is known as the Maxwell-Boltzmann distribution. At higher temperatures, the particles are distributed across a wider range of speeds.

The Maxwell-Boltzmann Distribution



Kinetic energy

The kinetic theory provides the relationship for the average kinetic energy of a single particle and temperature of the gas:

$$\frac{1}{2}m\overline{v^2} = \frac{3}{2}kT$$

The left hand side of the equation is the kinetic energy of a single particle. The Boltzmann constant, k is $1.38 \times 10^{-23} \text{JK}^{-1}$ and the temperature, T , must be on the absolute (Kelvin) scale. This simple equation gives a whole new explanation for temperature.

The temperature of the gas is directly dependent on the kinetic energy of the individual particles. This equation tells us that, for every 1K increase in temperature, the average kinetic energy of the particle increases by roughly $2 \times 10^{-23}\text{J}$. This is true for any gas particle: atom or molecule. If you mix Helium atoms with Carbon dioxide molecules, each of the two different particles will have the same average kinetic energy. Of course, the masses are quite different, so one must be travelling much faster on average.

Exam Hint: The average kinetic energy for each particle in a mixture of gases at the same temperature is the SAME. Hydrogen molecules will have the same average kinetic energy as oxygen molecules when mixed. As their masses are different, they will have different mean-squared-velocities.

Example 3

1. Calculate the rms velocity of a Helium atom at room temperature.
2. Which molecules will have more kinetic energy in a mixture of nitrogen and hydrogen molecules at the same temperature?
3. On average, will the hydrogen or nitrogen molecules be travelling fastest?

Answers

$$1. \text{ Rearrange } \frac{1}{2}m\overline{v^2} = \frac{3}{2}kT \text{ to give } \overline{v^2} = \frac{3kT}{m}$$

$$= \frac{3 \times 1.38 \times 10^{-23} \text{ JK}^{-1} \times 293 \text{ K}}{4 \times 1.661 \times 10^{-27} \text{ kg}}$$

$$\text{This gives } \overline{v^2} = 1.83 \times 10^6 \text{ m}^2 \text{ s}^{-2}.$$

$$\text{So } \sqrt{\overline{v^2}} = 1.35 \times 10^3 \text{ ms}^{-1} \text{ or } 1.4 \text{ km/s.}$$

2. As kinetic energy of a single molecule/ atom is given by $\frac{3}{2}kT$, the nitrogen and hydrogen molecules will have the same average kinetic energy.
3. The mass of the nitrogen molecules is fourteen times greater than the hydrogen molecules, so $\overline{v^2}$ must be fourteen times greater for the hydrogen molecules. $\sqrt{\overline{v^2}}$ for hydrogen molecules is 3.8x greater than for the nitrogen molecules.

Total Internal energy

Most of the energy within a gas is due to the particles moving about. In a molecular gas (like CO_2), the particles can have energy through spinning around or vibrating about their chemical bonds. However, most of the energy is kinetic energy due to the movement of the particles, which we call translational kinetic energy.

To calculate the total internal energy of a gas $U = \frac{1}{2}Nm\overline{v^2}$. This is simply the product of the number of molecules, N and the average kinetic energy of a single particle, $\frac{1}{2}m\overline{v^2}$.

Example 4

1. What is the total internal energy of 4 moles of oxygen molecules at 25°C ?
2. How many particles are there in a gas with 25,000 Joules total internal energy at 600°C ?

Answers

$$1. U = \frac{1}{2}Nm\overline{v^2} \text{ and } \frac{1}{2}m\overline{v^2} = \frac{3}{2}kT \text{ so } U = \frac{3}{2}NkT$$

$$T = 25 + 273.15 \text{ K} = 298.15 \text{ K}$$

$$N = 4 \times 6.022 \times 10^{23} = 2.4 \times 10^{24} \text{ particles}$$

$$U = \frac{3}{2} \times 2.4 \times 10^{24} \times 1.38 \times 10^{-23} \times 298.15 \text{ K} = 14.8 \text{ kJ}$$

$$2. U = \frac{3}{2}NkT \text{ becomes } 2U/3kT = N$$

$$N = \frac{2 \times 2.5 \times 10^4 \text{ J}}{(3 \times 1.38 \times 10^{-23} \times 873.15 \text{ K})} = 1.38 \times 10^{24} \text{ particles}$$

Practice Questions

1. Calculate the rms velocity of hydrogen molecules, oxygen molecules, argon atoms and radon atoms at room temperature.
2. Compare the average kinetic energy and rms velocity of helium atoms and carbon dioxide molecules in a mixture of gases.
3. A cylinder contains gas at 120 atmospheres at 20°C . A warning sign states that the maximum safe pressure is 240 atmospheres. What is the maximum safe temperature of the gas?
4. What is the average kinetic energy of a particle at 100°C ?
5. What is the total internal energy of 1.2 moles of steam at 177°C ?
6. The pressure inside a blow-up balloon is 1.3 atmospheres. The balloon is roughly 20cm across. Calculate the volume of the balloon by assuming it is spherical. How many moles of particles will be in the balloon at room temperature?
7. How many moles are there in 10.2kg of uranium hexafluoride gas? Uranium hexafluoride is a molecular gas, each molecule consists of one uranium atom and six fluorine atoms. Use the most common isotope for each element in your calculation.

7. 29.0 moles

6. 0.23 moles

5. 6.7KJ

4. $7.7 \times 10^{-21} \text{ J}$

3. 586K

2. Average kinetic energy of helium atoms and carbon dioxide molecules is the same. $\sqrt{\overline{v^2}}$ for helium is 3.3 times greater than for carbon dioxide molecules.

1. 1.9km/s, 480m/s, 430m/s, 180m/s

Answers

Acknowledgements:

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